THE MALAWI NATIONAL EXAMINATIONS BOARD

2022 MALAWI SCHOOL CERTIFICATE OF EDUCATION EXAMINATION

CHIEF EXAMINER'S REPORT

CHEMISTRY (M034)

PAPER I

A. GENERAL COMMENTS ON QUALITY OF THE PAPER AND PERFORMANCE OF CANDIDATES

The 2022 Chemistry paper 1 was generally fair as compared to the previous year. The level of difficulty was appropriate for the MSCE level. The language used was also appropriate for the MSCE candidate. The rubric of the paper was maintained quite well with adequate coverage of the syllabus. A wide range of scientific skills such as definition of terms, reasoning ability, drawing diagrams, interpretation of graphs and manipulation of figures were examined. Skills on ability to design and draw valid conclusions from results of an experiment were for instance, tested in questions **6 part c, 11 (a)** and (**b), 12** and **13**. The paper did not have typographical errors.

The overall performance of candidates was better than last year as number of candidates scoring zero has decreased compared to last year**.** Most candidates' responses displayed adequate coverage of the syllabus and good reasoning ability though some had problems to express themselves using scientific vocabulary and manipulation skills.

B. COMMENTS ON PERFORMANCE OF CANDIDATES ON SPECIFIC INDIVIDUAL QUESTIONS

SECTION A

Question 1a

The question demanded the candidates to name the isomers of alkanes **A** and **B** from the given information on a table and then explain why isomer **B** has a lower boiling point than **A.** This was a popular question. Most candidates attempted it and got it right.

However, some candidates did not get full marks because they failed to explain why the branched isomer has low boiling point due to poor mastery of content and their reasoning ability.

Question 1 b

Wanted candidates to give any three natural sources of water. Many attempted the question although most of them gave examples under one source. e.g. lake, river, sea. some candidates instead of giving sources of water, they were giving uses of water e.g. for washing, for domestic use etc.

Question 2 a

Part **a** required the candidates to give any two disadvantages of landfills as a way of disposing waste. Although this question was attempted by many candidates, most of them gave alternatives as independent answers as a result they lost marks.

Question 2 b

Candidates were asked to determine the temperature given the time from a graph of duration of heating against temperature, identify the dependent variable in the experiment and explain the relationship shown by the graph. Most candidates did well.

Question 3a

Candidates were required to explain the reason for storing phosphorous under water. A lot of candidates attempted it although most of them were not specific in their answers.

Question 3 b

Required candidates to give any four uses of phosphoric acid. Many candidates confused themselves with use of phosphorous as such they were giving uses of phosphorus and not phosphoric acid.

Question 3 c

Asked candidates to give any three differences between ionic and covalent compounds. Although it was attempted by many candidates, most of them were not differentiating the two as such some differences were not clear. For example, some wrote "ionic compounds conduct electricity while covalent does not. Ionic compounds have high melting points while covalent compounds are soluble in inorganic solvents".

Question 4 a

Part **a** demanded candidates to explain the depletion of ozone layer. It was attempted by many candidates although most of their answers were lacking support of chemical equations, hence lost some marks.

Question 4 a

Part **b** required candidates to give reasons as to why the bond between two atoms of bromine is called a non-polar covalent bond. A few candidates gave correct responses but most of them failed due to poor mastery of the content.

Question 5a

Candidates were asked to define the term "polymerization". The question was popular and many got it right except a few candidates who confused it with definition of isomers.

Question 5 b and c

Candidates were asked to complete the product of --------- given part of its equilibrium in a diagram. Many candidates had challenges in drawing the structure of the product and come up with the correct name. This happened possibly because the monomers were different.

Question 2 d

Required candidates to draw the structure of 2-methylpropan-2-ol. Many managed to draw but failed to put the hydroxyl group (OH) on the right positon. Some instead of putting –OH, they used ol in the structure, some put C=C while others wrote without the functional group, they were giving skeleton structures. They were required to write complete structures with all the hydrogen atoms included.

Question 6 a

Part **a** demanded candidates to identify compounds which are considered as "hydrocarbons" given different structures of organic compounds. Many candidates got it right.

Question 6 b

Required candidates to explain why one compound (**M)** has a higher boiling point than **L**. With reference to their given structures, many candidates attempted it but the reason was not clear and not scientific.

Question 6 c

Required candidates to describe how compound **N** can be distinguished from compound **L**. This question was attempted by most of the candidates but some gave contrary answers like use of bromine test. Other candidates were distinguishing using structures and results were not observable e.g. insoluble or soluble, positive or

negative. They ought to be specific by giving the observable results. Some could start with one test and give results of a different test.

Question 7 a

Required candidates to mention the type of oxide that is formed when metals react with oxygen. Many attempted and gave correct responses. However, some gave answers like metal oxide instead of giving the type of oxide.

Question 7 b

Demanded candidates to calculate the empirical formula of a compound given its percentage composition. A lot of candidates performed very well although some had challenges to find number of moles of H_2O. When writing an Empirical formula, some were writing commas e.g. $Na_{2,}C,O_3$, H_2O or $Na_2CO_3 + H_2O$.

Question 8 a

Asked candidates to define displacement reaction. Many candidates attempted it but their answers were not satisfactory e.g. most reactive displaces less reactive metal without talking about the state of the less reactive on i.e. in its solution.

Question 8 b

Wanted candidates to state any two applications of electroplating. The performance by most candidates was good. However, some were giving application of electrolysis.

Question 8 c (i)

Demanded candidates to explain how purity of a substance can be determined using standardized chromatography. This question was not popular and most candidates failed to explain properly.

Question 8 c (ii)

Required candidates to explain how a person smells perfume from a neighbouring friend. This was a popular question but most candidates gave funny answers. For example, some just wrote diffusion as an answer while others said my neighbour is poor so I couldn't smell perfume, this proved that the candidates did not master the content.

Question 9 a

Asked candidates to define the term "standard solution". Most candidates were ably defined the term although there were some who wrote as "it is the concentration of a known solution" instead of saying that "it is a solution of a known concentration (or a solution whose concentration is known)".

Question 9 b (i)

Required candidates to state the function of phenolphthalein solution in the acid- base reaction. Most candidates failed to give the exact function of phenolphthalein in relation to titration. Some could write "for identifying an acid or a base" instead of identifying the end point of the titration/ reaction.

Question 9 b (ii)

The question demanded candidates to write a balanced chemical equation for the reaction between sulphuric acid and sodium hydroxide. Most candidates lost marks because they failed to:

- predict the products of neutralization i.e. sodium sulphate and water
- write the correct molecular formula of sodium sulphate (Na_2SO_4)
- balance the equation correctly i.e. $2H_2SO_4 + 2\ NaOH \rightarrow 2Na_2SO_4 + 4H_2O$ instead of: $H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$

Question 9 b (iii)

Candidates were asked to calculate the molarity of the sulphuric acid solution. Most candidates used a dilution formula $C_1V_1 = C_2V_2$ instead of titration formula. $\frac{Ca \times Va}{na} = \frac{Cb \times Vb}{nb}$

In addition, failure to balance the equation led to failure in substitution.

Question 10 a (i)

The question required candidates to calculate the number of moles of water found in the hydrated copper sulphate crystals. Most candidates got the question correctly. However, some were adding instead of subtracting to get the mass of water. In addition, some did not even know the formula for number of moles $n = \frac{m}{Mr}$

Question 10 a (i)

In part **(ii),** candidates were asked to write a chemical formula of the hydrated copper sulphate, most candidates made an error in writing the formula i.e. $5H_2O.CuSO_4$ instead of $CuSO_4.5H_2O$. Some forgot to put a dot between the compound and water of crystallisation.

Question 10 b

Required candidates to give any two sources of alkanoates. A lot of candidates got the question correct. However, there was repetition of same concept especially natural sources as plants, animals, trees, flowers, leaves etc.

SECTION B

Question 11 a

The question was popular although a few got it right. Some failed because they were explaining ion exchange method or distillation is an experiment while other candidates could not include any procedure for an experiment in the explanation, could not clearly explain how the ions responsible for water hardness could be removed.

Many failed to get 5 out 5. Teachers should emphasize all methods when teaching

Question 11 b

The question was attempted by many candidates who got it right. However, some of the candidates who couldn't differentiate temperature change and Enthalpy change gave the procedure of the experiment that could be done rather than giving an explanation on endothermic reaction including the enthalpy change which was irrelevant. Others gave points for the procedure that were not logical; that is the first could come at the end. Skills of interpretation and experimental results should be emphasized after carrying out an experiment.

Question 12

Candidates were required to describe how butanol can be distinguished from 2-butanone using fehling test.

It seemed many candidates did not think "fehlings's test" could be conducted as a stand-alone test. Many candidates started with other tests such as solubility test, Brady's test, Bromine test, Tollens test and Acid test before describing the required Fehling test. Perhaps the knowledge of the flow diagram as a process for identifying unknown organic compounds confused the learners and they wrongly applied it here.

A great number of candidates were using general terms, specifically terms like alkanals and alkanones instead of butanal and 2-butanone, respectively. Some could not even mention the names butanal and 2-butanone in the whole of their answer, essentially they were answering the question of distinguishing alkanals from alkanones.
Other general words were “positive” and “negative” when describing the expected results of the fehling’s test.

Confusing “copper mirror” with “silver mirror”. Those candidates who attempted to describe fehling’s test, stated the expected results as silver mirror instead of copper mirror

Some students were answering the question as a question of separating a mixture of butanol and 2-butanone.

Question 13

The question asked candidates to explain on how an acid-base indicator can be prepared from hibiscus flowers.

It was a popular question and many got it right. However, some candidates

- Failed to reach proper conclusion e.g. testing the indicator in an acid or base and observe the colour changes. Some used litmus paper, universal, phenolphthalein (irrelevant because they were asked to describe how they can make an acid-base indicator not the already existing indicators).
- Many wrote irrelevant information instead of using the given hibiscus flowers, they used tomato leaves.

PAPER II

A. GENERAL COMMENTS ON QUALITY OF THE PAPER AND PERFORMANCE OF CANDIDATES

The 2022 Chemistry Paper II examination was fair. It covered syllabus adequately. Both the language and the level of difficulty of the test items were appropriate to the level of an MSCE candidate. The paper compared well with Chemistry Paper II from the past papers.

A wide range of skills were tested. These skills included instrument reading, observation, data recording, drawing, interpretation of graphs, ability to design experiments and draw valid conclusion and manipulation of figures.

Candidates performance was below average. Most candidates had problems to express themselves well using scientific vocabulary, misspelt some words and presented descriptive answers in a poor way. However, some candidates' responses to the questions indicated wide coverage of the syllabus.

B. COMMENTS ON THE PERFOMANCE OF CANDIDATES ON INDIVIDUAL QUESTIONS

Question 1

This question demanded candidates to describe the local preparation of ethanol with the aid of a well labelled diagram. It was a popular question and many candidates attempted it with a very good performance. Most of them were able to draw the correct diagram though some included a liebeg condenser and a thermometer.

The responses showed that the candidates had the following misconception:

- ethanol is produced during heating of the mixture of sugar, maize bran and water.
- carbon dioxide evaporates from the mixture and condense as ethanol on the other part of the apparatus.
- After evaporation has taken place, the remaining water in the boiling pot becomes ethanol.

Some candidates lost marks for:

- failing to include water in the mixture.
- writing equation for fermentation as $C_6H_{12}O_6 + H_2O \longrightarrow C_2H_5O + CO_2$
- incorrect labelling of the apparatus.
- failing to state that the mixture is filtered before it is heated.
- stating that the mixture is 'decanted' instead of filtered.
- stating that the mixture is 'fired' instead of heated.

Question 2

The question required candidates to explain the effect of surface area of reactants on reaction rate using digital balance, stop watch, 10g of powdered calcium carbonate, 10g of lamps of calcium carbonate, a conical flask, distilled water and dilute hydrochloric acid. The question was attempted by many candidates but poorly scored. Most candidates failed to describe steps in a sequential order to arrive at a valid conclusion. Most descriptions lacked an element of a fair test by not controlling variable e.g. using different volumes of HCl for powdered and lumps of calcium carbonate. In addition, most candidates lost marks for

- not controlling variables.
- including materials that were not mentioned in the question such as pestle and mortar, syringe, paper with a cross.
- not arriving at a valid conclusion.

- not stating that the time taken for powdered and lumps of calcium carbonate is compared.
- lack of sequential order of activities in the description. the responses indicated that the candidates had the following misconceptions;
- dissolving is a chemical reaction. Many wrote that powdered calcium carbonate dissolve faster that lumps of calcium carbonate hence surface area affects the rate of reaction.
- the surface area of the beaker affects the rate of reaction. Many talked of using a small beaker with small surface area and large beaker with large surface area.
- lumps are large hence large surface area and powder has small particles hence small surface area.
- water was a reactant in this experiment.
- Some candidates who missed the question and went on to give the general uses of the materials mentioned. Others simply gave the description that was on question 3.

Question 3

This question required candidates to conduct an experiment using sodium thiosalphate solution, filter paper, a top watch, black marker, a conical flask or beaker and hydrochloric acid of concentrations 0.1M, 0.3M and 0.5M.

Although many candidates attempted it, the performance was poor.

Most candidates failed to

- read time properly from the stopwatches.
- record time with appropriate units.
- use correct scale on the graph.
- misplacing variables on the axes
- come up with a smooth curve instead they drew a line of best.
- interpret the graph.

- state the relationship between concentration and rate of reaction properly. For example, they wrote 'concentration increases with increase in rate of reaction'.

Question 4

Candidates were asked to conduct an experiment using three test tubes, three beakers containing solutions of ionic compounds, a measuring cylinder, sodium hydroxide solution and distilled water. This was a popular question and most candidates had a good performance. However, some candidates failed the question because they;

- mistook colourless for white colour.
- wrote the whole compound instead of the cation present in the solution of the compound e.g. writing copper sulphate instead of copper ion.
- did not include the charge when they used chemical symbols for the ions present in the solutions e.g. writing Al instead of Al^{3+}.
- mismatched the colour of the cation identified e.g. green for Fe^{3+} instead of Fe^{2+}.
- wrote ammonium hydroxide or ammonium sulphate instead of ammonia solution.

C. RECOMMENDATIONS TO:

(i) Schools Should

- Procure enough materials for practical lessons.
- Ensure wide coverage of the syllabus, treating all topics equally; some students who got zeros and lower marks lacked reasoning skills and mastery of content.
- Encourage teachers to carry out practical activities and assess learners on the same.

- Encourage teachers to train students on how to read measuring instruments, research experiments e.g. in point form for descriptive and materials procedure, data collection, data analysis, conclusion for hands on experiments.
- Ensure that all science teachers have access to Chief Examiners' Reports.
- Conduct school and cluster-based INSETS for science teachers
- assist teachers with teaching and learning materials so that teachers conduct experiments during the teaching of practical lessons.
- Conduct school based supervision to improve on quality of delivery of instructions.

(ii) Malawi Institute of Education Should

- Revise text books that have errors

(iii) Ministry Should

- Recruit more Chemistry teachers to address the issue of being taught by unqualified teachers